

Johnny found a big brass button and set to work shining it on a piece of woolen cloth. "Isn't it bright?" he said, after working awhile. "Just like gold!"

He rubbed away again as hard as he could, then brushed the button across the back of his hand to wipe off some chalk dust. I had told him to put chalk on the cloth to brighten the button quicker. "Ow!" he cried, dropping the button. "Whats the matter?" "Its hot." "Hot!" echoed Mary, laying down her book. "How can it be hot?" "I don't know," said Johnny, "But it burned me."

"Nonsense!" replied Mary, picking up the button. "Its cold." "It may be now," Johnny admitted; "but it was hot- warm anyway." "What a silly boy! You imagined it." "I didn't," retorted Johnny. Seeing that they were likely to do as many old people have done, dispute about a matter that neither understood, I took the button and rubbed it smartly on my coat sleeve and then put it on Mary's cheek. "There!" exclaimed Johnny as Mary cried "Oh!" and put her hand on her face. "I shouldn't have thought your arm could make it so warm," she said. I rubbed on the button on the tablecloth and placed it once more against her cheek, saying: "It couldn't have been my arm that warmed it this time." "Of course not," observed Johnny.

"What did warm it?" Mary asked, her interest fully awakened. "Thats a good puzzle for you two to work at," I said. "Dont rub the button on the furniture for it might scratch that, but you can try anything else." Mary and Johnny worked for a long time and still were puzzled. "Maybe the heat comes from our fingers," Mary suggested at last. I put a stick through the button, so that it could be held without touching the hand. Then I rubbed it on the carpet and it was as hot as ever. "I guess it's the rubbing," said Johnny. "A good guess indeed, for that is precisely where the heat comes from," I replied. "The simple fact that heat comes from rubbing is perhaps enough for you to know about right now." "I thought heat always came from fire," said Mary, "or else from the sun." "There are other sources of heat," I replied; "our bodies for instance. We keep warm when out of the sunshine and away from the fire." "I never thought of that," said Mary. "Do you remember the day the masons were pouring water on a pile of quicklime to make mortar for the new house over the way? The lime hissed and crackled, sending up clouds of steam. I have a piece of quicklime here. See, when I pour water on it, how it drinks up the water and grows hot. I saw a wagon loaded with lime set on fire once by a shower of rain." "Fred told me about that, but I didn't believe him. Who'd expect fire from water?" "Get me a small piece of ice, I took from my cabinet a small bottle with a metal bead at the bottom. "Is it lead?" asked Johnny, when I showed it to him. "It is potassium," I said. "I'm going to set a little piece of it afire with the ice Mary brought. There ""Isn't it splendid!" cried Mary, as the metal flashed into flame. "You can do anything, can't you?" said Johnny admiringly. His confidence in my ability is something frightful. Really, if I were to tell him I could set the moon afire, I think he'd believe me. "No Johnny," I replied; "there are very few things I can do, as you will discover in time. But now, while we are talking of heat, let me show you another way of warming things. Please fetch me that old piece of iron in the garage, Mary, while Johnny brings me my hammer." When the materials were ready, I said: "Now watch me while I pound this piece of lead, and be ready to put your finger on it when I stop." "Does the pounding heat it?" "It does. I have seen a blacksmith take a piece of cold iron and hammer it on a cold anvil with a cold hammer until it was hot enough to set wood afire." "But we are a long way from



Johnny's button. Can you think of any other time you have seen things heated by rubbing?" "We rub our hands when they are cold," Mary said, seeing Fred go through these motions when coming

in from outdoors. "I've read of savages' making fire by rubbing sticks together," Fred continued.

"They have several ways of doing it- or rather, different savages have different ways.

One of the simplest is to rub on a stick in the groove in another, rubbing briskly and bearing down hard. There is a bit of soft pine board that I made to experiment with, the other day.

See! When I plow this stick up and down in the groove, the fine wood dust that gathers at the bottom begins to smoke a little and turns black. By working long enough and fast enough, I could set the dust on fire; but it is too tiresome when a match will do as well. We get our fire by rubbing, too, only we use something that kindles quicker than wood. A single scratch on some rough surface develops heat enough to light it "" "What is it?" Mary asked.

"Phosphorus. I have some in this bottle. You rub the button, Johnny, while I take some of the phosphorus out on the point of my knife. Now touch it with the button. See! It is hot enough to set the phosphorus afire. We might light our fires that way; but it is more convenient to put the phosphorus on the end of the stick, and mix it with something to keep it from catching fire too easily. All we have to do is to rub the phosphorus point against anything rough. The friction heats it and it takes fire. "Did you ever hear of the traveler who was stopped by some barbarous people who knew nothing of matches? They would not let him go through their country, and while they were debating whether to kill him or send him back he took a match from his pocket, struck against his boot, and lit it. To his surprise

The people ran off to the village. After a while the chief man came back humbly, bringing loads of presents. He begged the traveler to go on his way in peace." "What was the reason?"

They had seen him draw fire from his foot, as they thought, and were afraid that he was a god who might burn them all up if they offended him"

"In God's creation we are constantly surrounded with mysteries, many of which we are not able to understand yet, but Jesus promises to explain them in the heavenly home to those who remain obedient and faithful to Him," mom told us that evening